Data Profiling Report
NYC Yellow Taxi Trip Data (Jan 2024)

Generated: 2026-04-19 21:15
Rows analyzed: 2,964,624

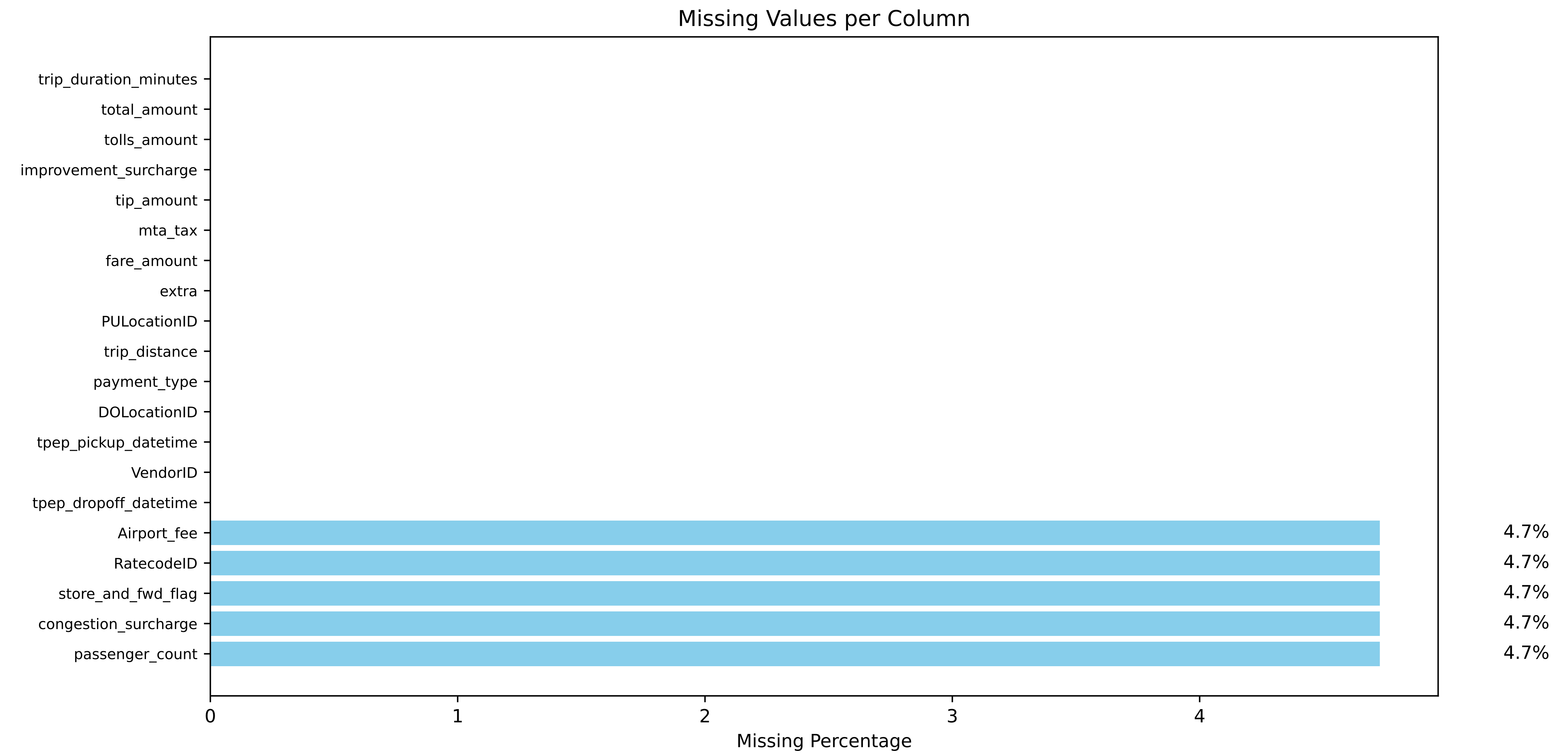
Assignment 2 – Big Data Analytics
NUST SEECS

Schema Description (all columns)

Column	Type	Sample Value
VendorID	int32	2
tpep_pickup_datetime	datetime64[us]	2024-01-01 00:57:55
tpep_dropoff_datetime	datetime64[us]	2024-01-01 01:17:43
passenger_count	float64	1.0
trip_distance	float64	1.72
RatecodeID	float64	1.0
store_and_fwd_flag	str	N
PULocationID	int32	186
DOLocationID	int32	79
payment_type	int64	2
fare_amount	float64	17.7
extra	float64	1.0
mta_tax	float64	0.5
tip_amount	float64	0.0
tolls_amount	float64	0.0
improvement_surcharge	float64	1.0
total_amount	float64	22.7
congestion_surcharge	float64	2.5
Airport_fee	float64	0.0
trip_duration_minutes	float64	19.8

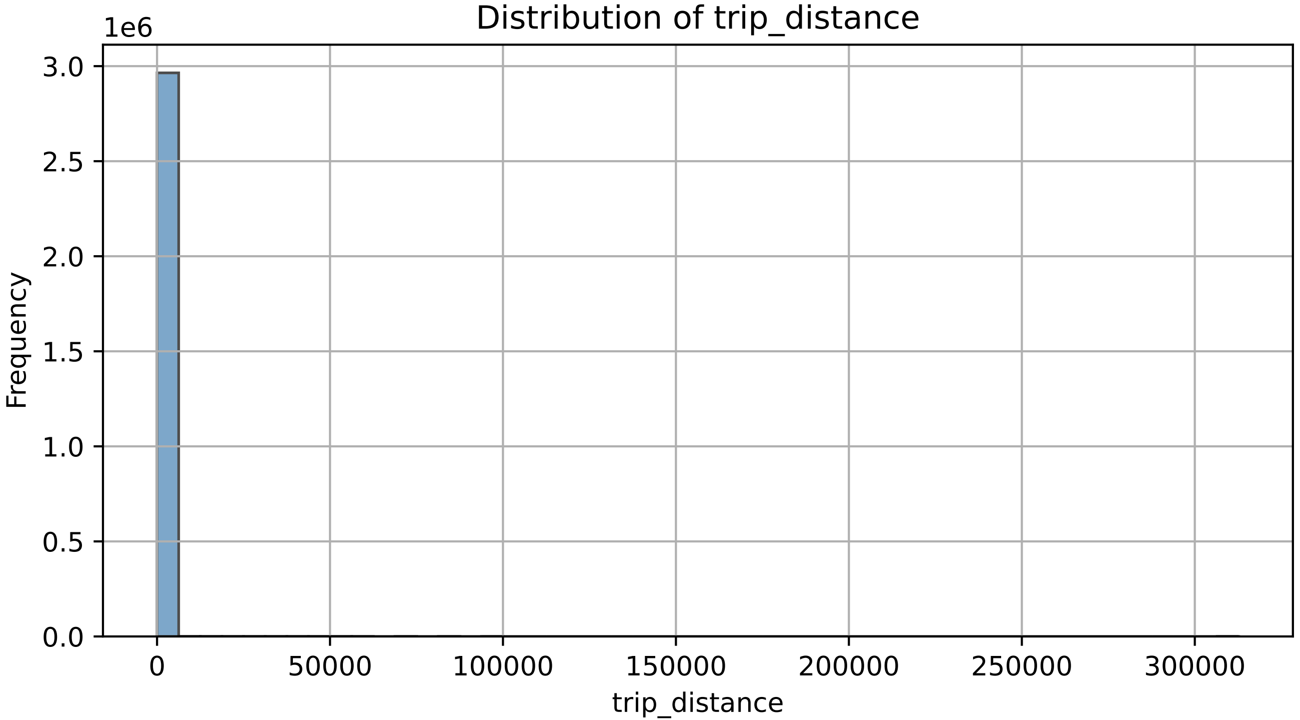
Missing Values per Column (count and percentage)

Column	Missing Count	Missing %
passenger_count	140162.0	4.73%
congestion_surcharge	140162.0	4.73%
store_and_fwd_flag	140162.0	4.73%
RatecodeID	140162.0	4.73%
Airport_fee	140162.0	4.73%
tpep_dropoff_datetime	0.0	0.00%
VendorID	0.0	0.00%
tpep_pickup_datetime	0.0	0.00%
DOLocationID	0.0	0.00%
payment_type	0.0	0.00%
trip_distance	0.0	0.00%
PULocationID	0.0	0.00%
extra	0.0	0.00%
fare_amount	0.0	0.00%
mta_tax	0.0	0.00%
tip_amount	0.0	0.00%
improvement_surcharge	0.0	0.00%
tolls_amount	0.0	0.00%
total_amount	0.0	0.00%
trip_duration_minutes	0.0	0.00%


Missing Values per Column
trip_duration_minutes
total_amount
tolls_amount
improvement_surcharge
tip_amount
mta_tax
fare_amount
extra
PULocationID
trip_distance
payment_type
DOLocationID
tpep_pickup_datetime
VendorID
tpep_dropoff_datetime
Airport_fee
RatecodeID
store_and_fwd_flag
congestion_surcharge
passenger_count
4.7%
4.7%
4.7%
4.7%
4.7%
0
1
2
3
4
Missing Percentage

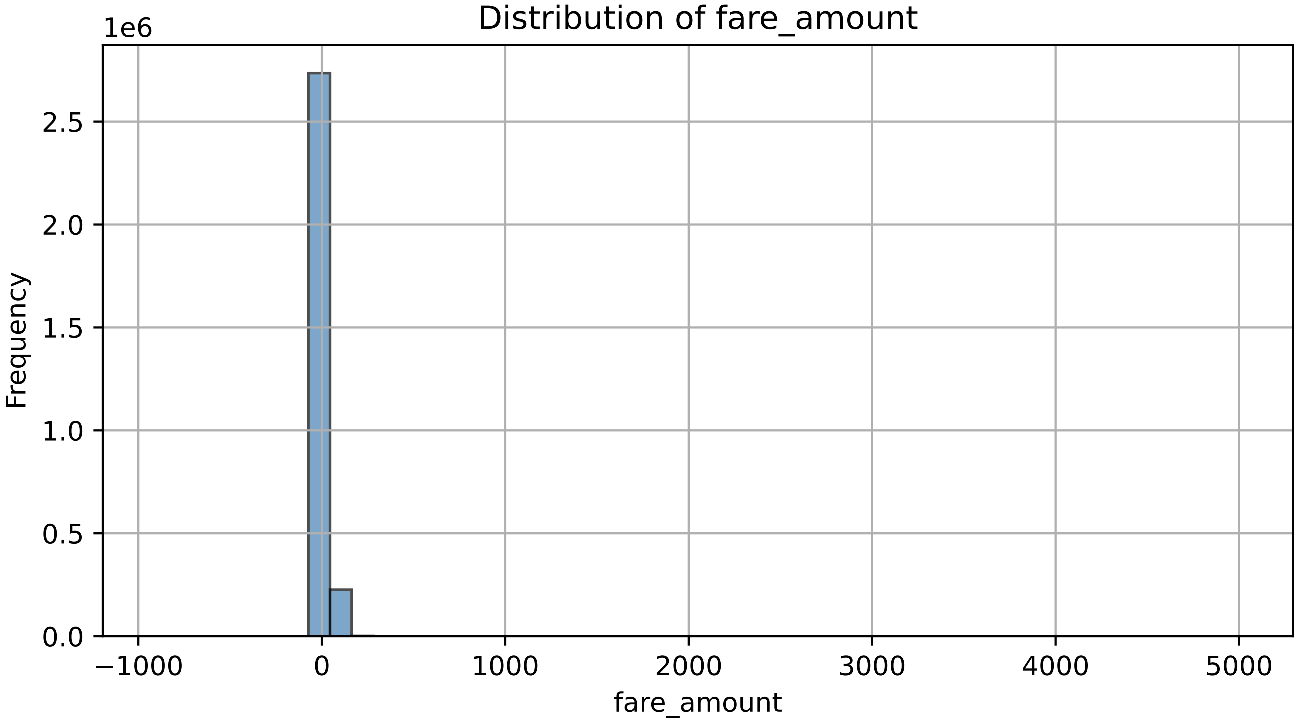
Statistical Summary – Numeric Columns (mean, median, std, min, max)

	mean	median	std	min	max
VendorID	1.75	2.0	0.43	1.0	6.0
passenger_count	1.34	1.0	0.85	0.0	9.0
trip_distance	3.65	1.68	225.46	0.0	312722.3
RatecodeID	2.07	1.0	9.82	1.0	99.0
PULocationID	166.02	162.0	63.62	1.0	265.0
DOLocationID	165.12	162.0	69.32	1.0	265.0
payment_type	1.16	1.0	0.58	0.0	4.0
fare_amount	18.18	12.8	18.95	-899.0	5000.0
extra	1.45	1.0	1.8	-7.5	14.25
mta_tax	0.48	0.5	0.12	-0.5	4.0
tip_amount	3.34	2.7	3.9	-80.0	428.0
tolls_amount	0.53	0.0	2.13	-80.0	115.92
improvement_surcharge	0.98	1.0	0.22	-1.0	1.0
total_amount	26.8	20.1	23.39	-900.0	5000.0
congestion_surcharge	2.26	2.5	0.82	-2.5	2.5
Airport_fee	0.14	0.0	0.49	-1.75	1.75
trip_duration_minutes	15.61	11.63	34.85	-13.57	9455.4



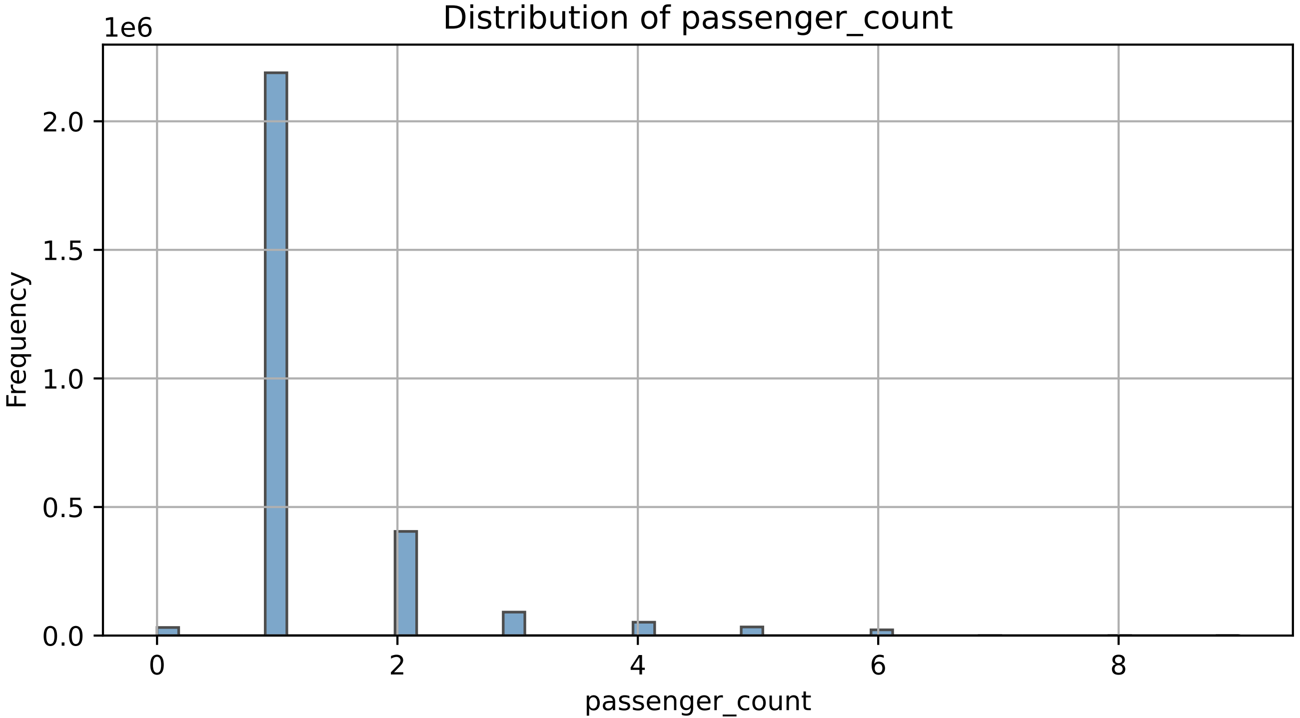
Distribution of trip_distance
1e6
3.0
2.5
2.0
1.5
1.0
0.5
0.0
Frequency
0
50000
100000
150000
200000
250000
300000
trip_distance

Interpretation for trip_distance:
Right-skewed. Most trips are under 5 miles, showing short urban trips dominate. Very few trips exceed 20 miles.

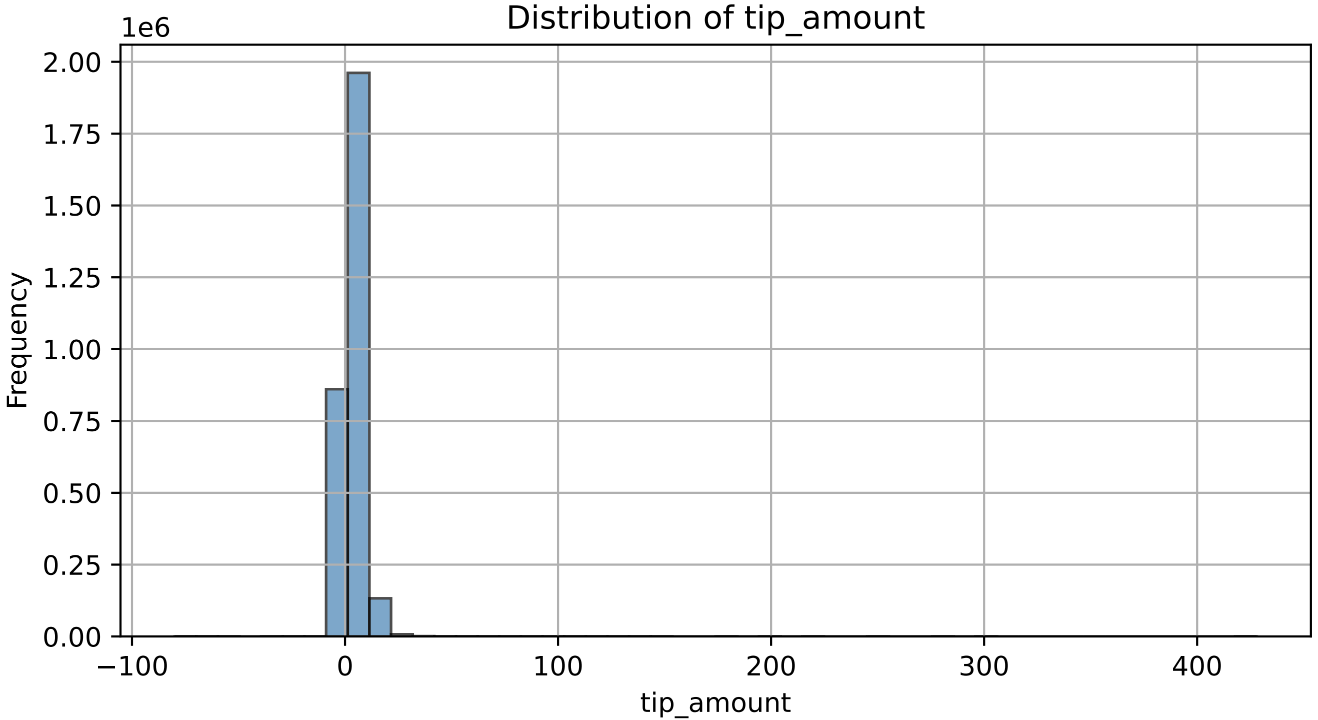


Distribution of fare_amount
1e6
2.5
2.0
1.5
1.0
0.5
0.0
Frequency
−1000
0
1000
2000
3000
4000
5000
fare_amount

Interpretation for fare_amount:
Right-skewed with a peak near 13. *Mostfaresare*10–20. *Faresabove*100 are outliers.

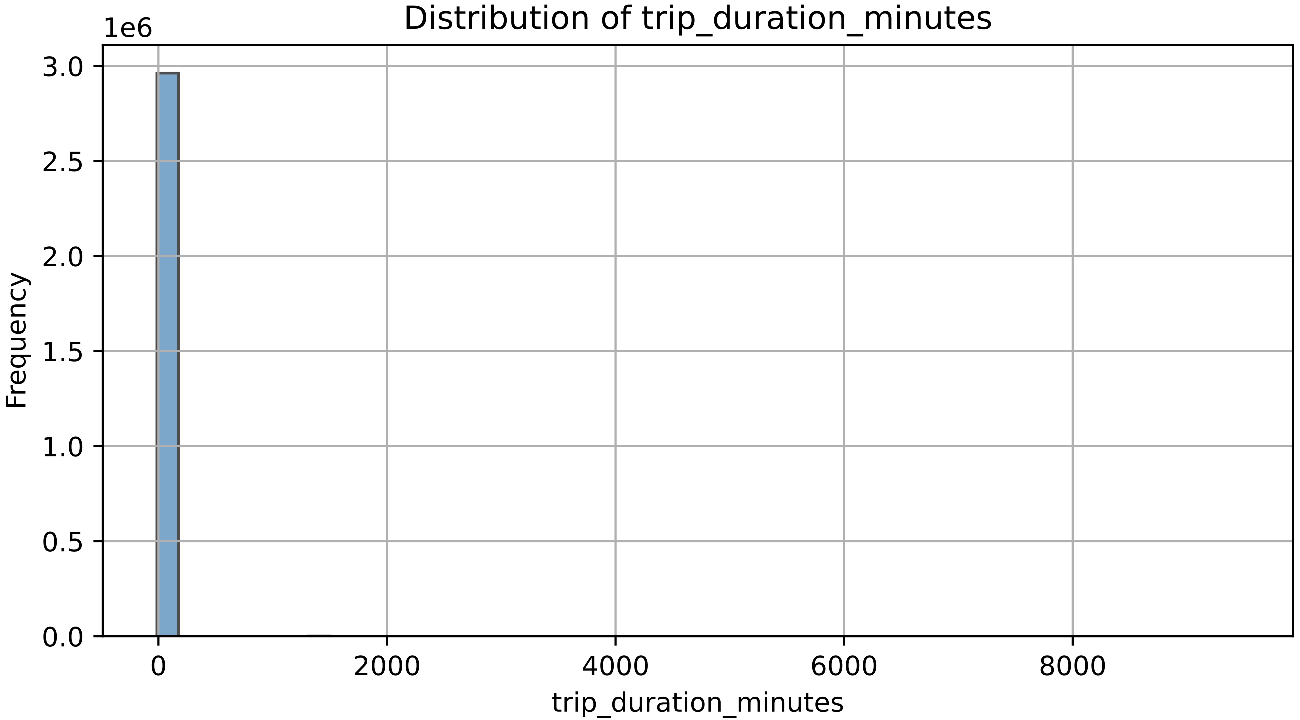

Distribution of passenger_count
1e6
2.0
1.5
1.0
0.5
0.0
Frequency
0
2
4
6
8
passenger_count

Interpretation for passenger_count:
Concentrated at 1 passenger (~70%). Trips with 2-3 passengers are less common. Zero or >4 passengers are errors.



Distribution of tip_amount
1e6
2.00
1.75
1.50
1.25
1.00
0.75
0.50
0.25
0.00
Frequency
−100
0
100
200
300
400
tip_amount

Interpretation for tip_amount:
Strongly right-skewed. Most common tip is 2–3. Many zero tips. High tips are rare.


Distribution of trip_duration_minutes
1e6
3.0
2.5
2.0
1.5
1.0
0.5
0.0
Frequency
0
2000
4000
6000
8000
trip_duration_minutes

Interpretation for trip_duration_minutes:
Right-skewed; typical trip lasts 10-20 minutes. Few trips exceed 60 minutes.

Identified Data Quality Issues (with counts/percentages):

- Duplicate rows: 0 (0.00%)
- Negative fare amount: 37448 (1.2632%)
- Zero passenger count: 31465 (1.06%)
- Passenger count non-integer: 140162 (4.73%)
- Trip distance outliers (IQR): 382745 (12.91%)
- Future pickup timestamps: 0 (0.0000%)
- Tip greater than fare: 41391 (1.3962%)

Proposed Cleaning Strategy (for Assignment 3)

Issue: Negative fare amount – 37448 (1.2632%) records.
Action: Replace negative values with the median fare amount for the same trip_distance bin (rounded to nearest 0.5 miles).
Justification: Median is robust against outliers; binning preserves distance-fare relationship.

Issue: Zero passenger count – 31465 (1.06%) records.
Action: Replace 0 with 1 (minimum plausible passenger).
Justification: A taxi trip cannot have zero passengers; imputing 1 is conservative.

Issue: Passenger count non-integer – 140162 (4.73%) records.
Action: Round to the nearest integer.
Justification: Passenger count must be a whole number.

Issue: Trip distance outliers – 382745 (12.91%) records.
Action: Cap outliers at the 99th percentile of trip_distance.
Justification: Extreme distances are likely erroneous; capping prevents distortion.

Issue: Duplicate rows – 0 (0.00%).
Action: Drop exact duplicates, keeping first occurrence.
Justification: Duplicates inflate counts and bias aggregations.

Issue: Future pickup timestamps – 0 (0.0000%) records.
Action: Replace with median pickup time for same hour-of-day and day-of-week.
Justification: Likely clock errors; imputing preserves temporal patterns.

Issue: Tip greater than fare – 41391 (1.3962%) records.
Action: Cap tip at fare amount (tip cannot exceed fare in normal circumstances).
Justification: Data entry errors; capping maintains logical consistency.

Missing values (any column): Numeric → median; categorical → mode; datetime → forward fill.
Justification: Simple, effective for warehouse ETL.